

Matrix

The numbers or functions are arranged in the form of a rectangular array, is known as matrix. If this array contains m rows and n columns, then it is called $m \times n$ matrix. The $m \times n$ numbers are called an element of the matrix.

Types of Matrices :-

- ① Row Matrix — A matrix having only one row and any number of columns, is called row matrix. i.e.
$$[-2 \ 3 \ 5]$$
- ② Column Matrix — A matrix having only one column and any number of rows, is called column matrix. i.e.
$$\begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix}$$
- ③ Square Matrix — A matrix in which number of rows is equal to number of columns, is called square matrix.
- ④ Null Matrix — A matrix in which each element is zero is called null matrix. i.e.
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$
- ⑤ Unit Matrix or Identity Matrix — A ^{square} matrix is called a unit matrix if the all elements of the principal diagonal are one and other elements are zero. i.e.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$
- ⑥ Transpose of a Matrix — If we interchange the rows and columns (corresponding), then the new matrix is known as transpose matrix.
If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 9 \\ 2 & 5 & 1 \end{bmatrix}$ then A^T or $A' = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 5 \\ 3 & 9 & 1 \end{bmatrix}$
- ⑦ Symmetric Matrix — A square matrix A is called symmetric matrix if $A' = A$, i.e.
$$\begin{bmatrix} a & h & g \end{bmatrix}$$

⑧ Skew-Symmetric Matrix:- A square matrix A , is called skew symmetric matrix if $A' = -A$ i.e.

$$\begin{bmatrix} 0 & -h & -g \\ h & 0 & -f \\ g & f & 0 \end{bmatrix}$$

⑨ Orthogonal Matrix:- A square matrix A is called orthogonal matrix if $A \cdot A' = I = A' \cdot A$

⑩ Singular & Non-Singular Matrix:- A square matrix A is called singular matrix if $|A| = 0$, and non-singular if $|A| \neq 0$

⑪ Conjugate of a Matrix:- we replace i by $-i$,
if $A = \begin{bmatrix} 1+i & 2i & 0 \\ -2 & 3-2i & i \end{bmatrix}$ then $\bar{A} = \begin{bmatrix} 1-i & -2i & 0 \\ i & 3+2i & -i \end{bmatrix}$

⑫ Matrix A^θ :- the transpose of the conjugate (or conjugate of the transpose) of the matrix A is called A^θ . i.e.
 $A^\theta = \overline{(A')}$ or $A^\theta = (\bar{A})'$

⑬ Unitary Matrix:- A square matrix A is said to unitary matrix if $A^\theta A = I$ or $AA^\theta = I$.

⑭ Hermitean & Skew Hermitean Matrix:- A square matrix A is said to be Hermitean if $A^\theta = A$, and skew Hermitean if $A^\theta = -A$.

⑮ Nilpotent Matrix:- A matrix A is called nilpotent matrix if $A^k = 0$ where k is the index of the nilpotent matrix. if $k=2$, then $A^2 = 0$.

⑯ Idempotent Matrix:- A matrix A is called idempotent if $A^2 = A$.

⑰ Involuntary Matrix:- A matrix A is called involuntary if $A^2 = I$

Que ① Show that any square matrix can be expressed as the sum of two matrices one symmetric and other skew symmetric.

Proof:-

$$A = \left(\frac{A+A'}{2} \right) + \left(\frac{A-A'}{2} \right) = \frac{\alpha}{2} + \frac{\beta}{2}$$

Now $\alpha' = (A+A')' = A' + (A')' = A' + A = \alpha$ i.e. symmetric

& $\beta' = (A-A')' = A' - (A')' = A' - A = -(A-A') = -\beta$ i.e. skew symmetric.

Que ② Write the given matrix as the sum of a symmetric and skew symmetric matrix.

$$A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 5 & 3 \\ -1 & 6 & 3 \end{bmatrix}$$

Sol:-

$$A' = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 5 & 6 \\ 4 & 3 & 3 \end{bmatrix}$$

$$A+A' = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 10 & 9 \\ 3 & 9 & 6 \end{bmatrix} \text{ --- ①} \quad \& \quad A-A' = \begin{bmatrix} 0 & 4 & 5 \\ -4 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix} \text{ --- ②}$$

$$\text{①} + \text{②} \Rightarrow A + A' + A - A' = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 10 & 9 \\ 3 & 9 & 6 \end{bmatrix} + \begin{bmatrix} 0 & 4 & 5 \\ -4 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix}$$

$$\Rightarrow A = \frac{1}{2} \begin{bmatrix} 2 & 0 & 3 \\ 0 & 10 & 9 \\ 3 & 9 & 6 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 & 4 & 5 \\ -4 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix}$$

symmetric skew symmetric.

Que ③ :- Show that $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ is orthogonal.

Sol:-

$$A' = \frac{1}{3} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$$

$$A \cdot A' = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Que ④ :- Determine the value of α, β, γ where $A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal.

Sol:-

$$A' = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}$$

given that A is orthogonal i.e. $AA' = I$, hence

$$\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \end{bmatrix} \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 - \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Que ⑤ :- Show that $A = \begin{bmatrix} 1 & 1-i & 2 \\ 1+i & 3 & i \\ 2 & -i & 0 \end{bmatrix}$ is Hermitian Matrix.

Sol :- $\bar{A} = \begin{bmatrix} 1 & 1+i & 2 \\ 1-i & 3 & -i \\ 2 & i & 0 \end{bmatrix}$ Now $(\bar{A})' = \begin{bmatrix} 1 & 1-i & 2 \\ 1+i & 3 & i \\ 2 & -i & 0 \end{bmatrix} \Rightarrow \underline{A^{\theta} = A}$

Que ⑥ :- Show that $A = \begin{bmatrix} -i & 3+2i & -2-i \\ -3+2i & 0 & 3-4i \\ 2-i & -3-4i & -2i \end{bmatrix}$ is Skew-Hermitian Matrix.

Sol :- $\bar{A} = \begin{bmatrix} i & 3-2i & -2+i \\ -3-2i & 0 & 3+4i \\ 2+i & -3+4i & 2i \end{bmatrix}$ Now $(\bar{A})' = \begin{bmatrix} i & -3-2i & 2+i \\ 3-2i & 0 & 3+4i \\ -2+i & 3+4i & 2i \end{bmatrix} = -\begin{bmatrix} -i & 3+2i & -2-i \\ -3+2i & 0 & 3-4i \\ 2-i & -3-4i & -2i \end{bmatrix}$
or $\underline{A^{\theta} = -A}$

Que ⑦ :- Find the nature of the following matrices -
 $A+A^{\theta}$, AA^{θ} , $A-A^{\theta}$

Sol :- $(A+A^{\theta})^{\theta} = A^{\theta} + (A^{\theta})^{\theta} = A^{\theta} + A \rightarrow \text{Hermitian}$
 $(AA^{\theta})^{\theta} = A^{\theta} (A^{\theta})^{\theta} = A^{\theta} A = AA^{\theta} \rightarrow \text{Hermitian}$
 $(A-A^{\theta})^{\theta} = A^{\theta} - (A^{\theta})^{\theta} = A^{\theta} - A = -(A-A^{\theta}) \rightarrow \text{Skew-Hermitian}$

Que ⑧ :- If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, show that A^*A is Hermitian matrix, where A^* is the conjugate transpose of A .

Sol :- $A' = \begin{bmatrix} 2+i & -5 \\ 3 & i \\ -1+3i & 4-2i \end{bmatrix}$ & $(\bar{A}') = \begin{bmatrix} 2-i & -5 \\ 3 & -i \\ -1-3i & 4+2i \end{bmatrix} = A^*$

Now $A^*A = \begin{bmatrix} 2-i & -5 \\ 3 & -5 \\ -1-3i & 4+2i \end{bmatrix} \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix} = \begin{bmatrix} 30 & 6-8i & -19+17i \\ 6+8i & 10 & -5+5i \\ -19-17i & -5-5i & 30 \end{bmatrix} = B$

Now $\bar{B} = \begin{bmatrix} 30 & 6+8i & -19-17i \\ 6-8i & 10 & -5-5i \\ -19+17i & -5+5i & 30 \end{bmatrix} \Rightarrow B^{\theta} = (\bar{B})' = \begin{bmatrix} 30 & 6-8i & -19+17i \\ 6+8i & 10 & -5+5i \\ -19-17i & -5-5i & 30 \end{bmatrix}$

$\therefore B^{\theta} = B$ i.e. $B = A^{\theta}A$ is Hermitian matrix.

Que 9 Show that every square matrix can be expressed as $R + iS$, where R & S are Hermitian Matrix.

Sol:-
$$A = \frac{1}{2} (A + A^{\theta}) + i \cdot \frac{1}{2i} (A - A^{\theta}) = R + iS$$

 where $R = \frac{1}{2} (A + A^{\theta})$ & $S = \frac{1}{2i} (A - A^{\theta})$

Now $R^{\theta} = \frac{1}{2} (A + A^{\theta})^{\theta} = \frac{1}{2} [A^{\theta} + (A^{\theta})^{\theta}] = \frac{1}{2} (A^{\theta} + A) = \frac{1}{2} (A + A^{\theta}) = R$
 & $S^{\theta} = \left[\frac{1}{2i} (A - A^{\theta}) \right]^{\theta} = \frac{1}{-2i} [A^{\theta} - (A^{\theta})^{\theta}] = \frac{1}{-2i} (A^{\theta} - A) = \frac{1}{2i} (A - A^{\theta}) = S$
 i.e. R & S both are Hermitian matrix.

Que 10 :- Show that any square matrix can be expressed as the sum of two matrices one Hermitian and other skew Hermitian.

Sol:-
$$A = \frac{1}{2} (A + A^{\theta}) + \frac{1}{2} (A - A^{\theta}) = P + Q.$$

 where $P = \frac{1}{2} (A + A^{\theta})$ & $Q = \frac{1}{2} (A - A^{\theta})$

Now $P^{\theta} = \frac{1}{2} (A + A^{\theta})^{\theta} = \frac{1}{2} [A^{\theta} + (A^{\theta})^{\theta}] = \frac{1}{2} (A^{\theta} + A) = P$
 and $Q^{\theta} = \frac{1}{2} (A - A^{\theta})^{\theta} = \frac{1}{2} [A^{\theta} - (A^{\theta})^{\theta}] = \frac{1}{2} (A^{\theta} - A) = -\frac{1}{2} (A - A^{\theta}) = -Q$
 i.e. $P^{\theta} = P$ & $Q^{\theta} = -Q$. hence P is Hermitian

and Q is skew-Hermitian.

Que 11 :- Expressed the matrix $A = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix}$ as the sum

of Hermitian and skew-Hermitian matrix.

Sol:- Now $\bar{A} = \begin{bmatrix} 1-i & 2 & 5+5i \\ -2i & 2-i & 4-2i \\ -1-i & -4 & 7 \end{bmatrix}$ $\therefore A^{\theta} = (\bar{A})' = \begin{bmatrix} 1-i & -2i & -1-i \\ 2 & 2-i & -4 \\ 5+5i & 4-2i & 7 \end{bmatrix}$

Now $P = \frac{1}{2} (A + A^{\theta}) = \frac{1}{2} \begin{bmatrix} 2 & 2-2i & 4-6i \\ 2+2i & 4 & 2i \\ 4+6i & -2i & 14 \end{bmatrix} = \begin{bmatrix} 1 & 1-i & 2-3i \\ 1+i & 2 & i \\ 2+3i & -i & 7 \end{bmatrix}$

& $Q = \frac{1}{2} (A - A^{\theta}) = \begin{bmatrix} i & 1+i & 3-2i \\ -1+i & i & 4+i \\ -3-2i & -4+i & 0 \end{bmatrix}$

(5)

Hence $A = P + Q = \begin{bmatrix} 1 & 1-i & 2-3i \\ 1+i & 2 & i \\ 2+3i & -i & 7 \end{bmatrix} + \begin{bmatrix} i & 1+i & 3-2i \\ -1+i & i & 4+i \\ -3-2i & -4+i & 0 \end{bmatrix}$

Que (12) :- Express $A = \begin{bmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{bmatrix}$ as the sum of Hermitian and skew-Hermitian matrix.

Que (13) :- If $A = \begin{bmatrix} 3 & 5+2i & -3 \\ 5-2i & 7 & 4i \\ -3 & -4i & 5 \end{bmatrix}$ show that A is Hermitian and iA is a skew Hermitian.

Que (14) :- Prove that the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary.

Sol :- $A^{\theta} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$

Now $A^{\theta} A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

Que (15) :- Define a unitary matrix. If $N = \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix}$ is a matrix, then show that $(I-N)(I+N)^{-1}$ is a unitary matrix, where I is an identity matrix.

Sol :- Now $I - N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$

and $I + N = \begin{bmatrix} 1 & 1+2i \\ -1+2i & 1 \end{bmatrix}$

again Now $(I+N)^{-1} = \frac{\text{adj}(I+N)}{|I+N|} = \frac{1}{6} \begin{bmatrix} 1 & -1-2i \\ 1-2i & 1 \end{bmatrix}$

Now $(I-N)(I+N)^{-1} = \frac{1}{6} \begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix} = B \text{ (suppose)}$

Now $B^{\theta} = \frac{1}{6} \begin{bmatrix} -4 & 2+4i \\ -2-4i & -4 \end{bmatrix}$

$\therefore B^{\theta} \cdot B = \frac{1}{6} \begin{bmatrix} -4 & 2+4i \\ -2-4i & -4 \end{bmatrix} \cdot \frac{1}{6} \begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix} = \frac{1}{36} \begin{bmatrix} 36 & 0 \\ 0 & 36 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

Que (16) :- Show that $A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ -i & -1 \end{bmatrix}$ is unitary matrix.

Que (17) :- Show that $V = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$ is a unitary matrix

where ω is complex cube root of unity.

⑥

Que (18) :- Using elementary transformation find inverse of the matrix -

(i) $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$

(iv) $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

Ans :- (i) $\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$

(ii) $\frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -8 & 6 & -2 \\ 5 & -3 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$

(iv) $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -3 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{bmatrix}$

Que (19) :- Find the rank of the matrix by triangular form or Echelon form

(i) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 2 & 3 & -4 \\ -2 & 3 & 7 & -1 \\ 1 & 9 & 16 & -13 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$

(iv) $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$

(v) $\begin{bmatrix} 3 & 4 & 1 & 1 \\ 2 & 4 & 3 & 6 \\ -1 & -2 & 6 & 4 \\ 1 & -1 & 2 & 6 \end{bmatrix}$

(vi) $\begin{bmatrix} 2 & -4 & 3 & 1 & 0 \\ 4 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$

(vii) $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$

Ans :- (i) 2, (ii) 2, (iii) 2, (iv) 3, (v) 4, (vi) 3, (vii) 3

Rank of a Matrix :- The rank of a matrix is said to be r if

- It has at least one non-zero minor of order r .
- Every minor of order higher than r is zero.

Que (20) :- For which value of λ the rank of the matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ \lambda & 13 & 10 \end{bmatrix}$ is 2.

Sol :- 1st Method :-

Since the rank of the matrix A is 2, hence the minor of order three must be zero.

i.e. $\begin{vmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ \lambda & 13 & 10 \end{vmatrix} = 0 \Rightarrow 1(30-26) + \lambda(10-12) = 0 \Rightarrow 4 - 2\lambda = 0 \Rightarrow \boxed{\lambda = 2}$

2nd Method :- We convert the matrix into Echelon Form.

$A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ \lambda & 13 & 10 \end{bmatrix}$

$R_3 \rightarrow R_3 - \lambda R_1$

$\sim \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ 0 & 13-5\lambda & 10-4\lambda \end{bmatrix}$

$R_3 \rightarrow 3R_3 - (13-5\lambda)R_2$

$\begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ 0 & 0 & 3(10-4\lambda) - 2(13-5\lambda) \end{bmatrix}$

Since $\rho(A) = 2$ hence

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Normal Form :- (i) $[I_r]$, (ii) $[I_r \ 0]$, (iii) $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \rightarrow$ canonical form.

Que (21) :- Find the rank of the matrix by reducing into normal form :-

$$(i) \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 2 & 3 & -2 & 4 \\ 3 & -2 & 1 & 2 \\ 3 & 2 & 3 & 4 \\ -2 & 4 & 0 & 5 \end{bmatrix}$$

Ans :- (i) Rank = 3 (ii) Rank = 3 (iii) Rank = 3

Que (22) :- Find the non-singular matrix P and Q such that PAQ is normal form.

$$(i) \begin{bmatrix} 1 & 1 & 2 \\ 2 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}, (ii) \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}, (iii) \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$$

Que (23) :- If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, Find two non-singular matrix P and Q, such that $PAQ = I$. Hence find A^{-1} .

Sol :-

$$A_{3 \times 3} = I_3 A I_3$$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_2 \rightarrow -C_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ -2 & 3 & 0 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & -3 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - C_2$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

i.e.

$$I_3 = P A Q$$

Now $PAQ = I$

Pre-multiply by P^{-1}

$$\therefore P^{-1} PAQ = P^{-1} I$$

$$\Rightarrow AQ = P^{-1}$$

Post multiply by P

$$AQP = P^{-1}P$$

$$\Rightarrow AQP = I$$

$$\Rightarrow QP = A^{-1}$$

$$\text{or } \boxed{A^{-1} = QP}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

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Ans

Solution of Simultaneous Equations:

$$\begin{array}{r} x + y = 4 \\ x - y = 2 \\ \hline \text{Unique Sol.} \end{array}$$

$$\begin{array}{r} x + y = 4 \\ 2x + 2y = 8 \\ \hline \text{Infinite Sol.} \end{array}$$

$$\begin{array}{r} x + y = 4 \\ 2x + 2y = 5 \\ \hline \text{No. Sol.} \end{array}$$

Consistency of a system of linear equations. —

$$\left. \begin{array}{l} a_1 x + b_1 y + c_1 z = d_1 \\ a_2 x + b_2 y + c_2 z = d_2 \\ a_3 x + b_3 y + c_3 z = d_3 \end{array} \right\} \Rightarrow \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$\boxed{A X = B}$$

Now $E = [A : B]$ is called augmented matrix.

(a) Consistent equation :— If $\rho(A) = \rho(E)$

(i) Unique Sol. $\rightarrow \rho(A) = \rho(E) = n$ (no. of unknown)

(ii) Infinite Sol. $\rightarrow \rho(A) = \rho(E) = r < n$

(b) In-consistent equation :— If $\rho(A) \neq \rho(E)$ i.e. no-solution.

Note — The system of equations will have a unique solution if and only if the coefficient matrix is non-singular.

i.e. for unique sol $\rightarrow |A| \neq 0$
and infinite sol $\rightarrow |A| = 0$

Que (21) :— Discuss the consistency of the following system of equations —

$$2x + 3y + 4z = 11, \quad x + 5y + 7z = 15, \quad 3x + 11y + 13z = 25$$

if found consistent solve it.

Sol :— $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 7 \\ 3 & 11 & 13 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 11 \\ 15 \\ 25 \end{bmatrix}$

The augmented matrix $C = [A : B]$

$$C = \begin{bmatrix} 2 & 3 & 4 & : & 11 \\ 1 & 5 & 7 & : & 15 \\ 3 & 11 & 13 & : & 25 \end{bmatrix}$$

$R_1 \leftrightarrow R_2$

$$\begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 2 & 3 & 4 & : & 11 \\ 3 & 11 & 13 & : & 25 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 0 & -7 & -10 & : & -19 \\ 0 & -4 & -8 & : & -20 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_3$$

$$R_3 \rightarrow R_3 + 4R_2$$

$$\begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 0 & 1 & 6 & : & 21 \\ 0 & -4 & -8 & : & -20 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 0 & 1 & 6 & : & 21 \\ 0 & 0 & 16 & : & 64 \end{bmatrix}$$

Now

$$A X = B$$

$$\begin{bmatrix} 1 & 5 & 7 \\ 0 & 1 & 6 \\ 0 & 0 & 16 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 64 \end{bmatrix}$$

$$\Rightarrow x + 5y + 7z = 15$$

$$y + 6z = 21$$

$$16z = 64 \Rightarrow z = 4, y = -3 \text{ \& } x = 2 \quad \underline{\text{Ans}}$$

Que (22) :- Show that the equations :- $2x + 6y = -11$,

$$6x + 20y - 6z = -3, \quad 6y - 18z = -1, \quad \text{are not consistent.}$$

Sol :-

$$A = \begin{bmatrix} 2 & 6 & 0 \\ 6 & 20 & -6 \\ 0 & 6 & -18 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} -11 \\ -3 \\ -1 \end{bmatrix}$$

$$\text{Now } C \cong [A : B] = \begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 6 & 20 & -6 & : & -3 \\ 0 & 6 & -18 & : & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$\sim \begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 0 & 2 & -6 & : & 30 \\ 0 & 6 & -18 & : & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 0 & 2 & -6 & : & 30 \\ 0 & 0 & 0 & : & -91 \end{bmatrix}$$

$$\rho(A) \neq \rho(C) \quad \text{i.e. inconsistent.}$$

Que (23) :- Test for consistency the following system of equations and if consistent, solve it.

$$(i) \quad 3x + 3y + 2z = 1, \quad x + 2y = 4, \quad 10y + 3z = -2, \quad 2x - 3y - z = 5$$

$$(ii) \quad x_1 + 2x_2 - x_3 = 3, \quad 3x_1 - x_2 + 2x_3 = 1, \quad 2x_1 - 2x_2 + 3x_3 = 2, \quad x_1 - x_2 + x_3 = -1$$

$$(iii) \quad 2x - y + 3z = 8, \quad -x + 2y + z = 4, \quad 3x + y - 4z = 0$$

$$(iv) \quad x + 2y + 2z = 1, \quad 2x + y + z = -2, \quad 3x + 2y + 2z = 3, \quad y + z = 0$$

$$\text{Ans :- } (i) \quad x = 2, y = 1, z = -4$$

$$(ii) \quad x_1 = -1, x_2 = 4, x_3 = 4$$

$$(iii) \quad x = 1, y = 2, z = 2$$

(9)

Ques (24) :- Solve the equation using matrix method

$$x_1 + 3x_2 + 2x_3 = 0, \quad 2x_1 - x_2 + 3x_3 = 0$$

$$3x_1 - 5x_2 + 4x_3 = 0, \quad x_1 + 17x_2 + 4x_3 = 0$$

Sol :-

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & -1 & 3 \\ 3 & -5 & 4 \\ 1 & 17 & 4 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Now } C \cong [A : B] = \begin{bmatrix} 1 & 3 & 2 & : & 0 \\ 2 & -1 & 3 & : & 0 \\ 3 & -5 & 4 & : & 0 \\ 1 & 17 & 4 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1, \quad R_4 \rightarrow R_4 - R_1$$

$$\sim \begin{bmatrix} 1 & 3 & 2 & : & 0 \\ 0 & -7 & -1 & : & 0 \\ 0 & -14 & -2 & : & 0 \\ 0 & 14 & 2 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2, \quad R_4 \rightarrow R_4 + 2R_2$$

$$\sim \begin{bmatrix} 1 & 3 & 2 & : & 0 \\ 0 & -7 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\rho(A) = \rho(C) = 2 \quad (< \text{no. of unknowns i.e. } 3)$$

$\therefore$ i.e. non trivial sol. or infinite sol.

$$\therefore A X = B$$
$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & -7 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x_1 + 3x_2 + 2x_3 &= 0 \\ -7x_2 - x_3 &= 0 \end{aligned}$$

$$\text{Let } x_2 = k \quad \therefore x_3 = -7k$$

$$\therefore x_1 + 3k - 14k = 0$$

$$\Rightarrow x_1 = 11k$$

(10)

Ans

Que (25) :- Determine for what value of λ and μ the following equations have, (i) no sol. (ii) Unique sol. (iii) infinite sol.

$$x+y+z=6, \quad x+2y+3z=10, \quad x+2y+\lambda z=\mu$$

Sol :- $C = [A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix}$

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & \lambda-1 & : & \mu-6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda-3 & : & \mu-10 \end{bmatrix}$$

(i) No. solution if $\rho(A) \neq \rho(C)$ i.e. $\lambda-3=0$ & $\mu-10 \neq 0$
or $\lambda=3$, $\mu \neq 10$

(ii) Unique sol. if $\rho(A) = \rho(C)$ i.e. $\lambda-3 \neq 0$ & μ can take any value.

(iii) Infinite sol if $\rho(A) = \rho(C) = 2 (< 3)$ i.e. $\lambda-3=0$ & $\mu-10=0$
or $\lambda=3$ & $\mu=10$.

Que (26) :- Find the value of b such that the system of equations

$$2x+y+2z=0, \quad x+y+3z=0, \quad 4x+3y+bz=0$$

has non-trivial solution.

Sol :-

$$C \approx [A : B] = \begin{bmatrix} 2 & 1 & 2 & : & 0 \\ 1 & 1 & 3 & : & 0 \\ 4 & 3 & b & : & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 & 3 & : & 0 \\ 2 & 1 & 2 & : & 0 \\ 4 & 3 & b & : & 0 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{matrix}} \begin{bmatrix} 1 & 1 & 3 & : & 0 \\ 0 & -1 & -4 & : & 0 \\ 0 & -1 & b-12 & : & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 1 & 3 & : & 0 \\ 0 & -1 & -4 & : & 0 \\ 0 & 0 & b-8 & : & 0 \end{bmatrix}$$

for non trivial sol or infinite sol $\rho(A) = \rho(C) = 2 (< \text{no. of unknown})$
hence $b-8=0$ or $b=8$.

And Method :- For non trivial solution $|A| = 0$

$$\Rightarrow \begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & b \end{vmatrix} = 0$$

$$\Rightarrow 2(b-9) - 1(b-6) + 4(3-2) = 0$$

$$\Rightarrow 2b - 18 - b + 6 + 4 = 0$$

$$\Rightarrow b - 8 = 0$$

Ans

Que (27) :- Find the value of k such that the following equations have unique solution.

$$\lambda x + 2y - 2z - 1 = 0, \quad 4x + 2\lambda y - 3z - 2 = 0, \quad 6x + 6y + \lambda z - 3 = 0$$

Sol :- For unique sol. $|A| \neq 0$

$$\Rightarrow \begin{vmatrix} \lambda & 2 & -2 \\ 4 & 2\lambda & -1 \\ 6 & 6 & \lambda \end{vmatrix} \neq 0 \Rightarrow \lambda^3 + 11\lambda - 30 \neq 0$$

$$(\lambda - 2)(\lambda^2 + 2\lambda + 15) \neq 0$$

$$\Rightarrow \lambda \neq 2. \quad \text{Ans}$$

Que (28) :- Find the values of k such that the system of eqn^s has non-trivial solution.

$$x + ky + 3z = 0, \quad 4x + 3y + kz = 0, \quad 2x + y + 2z = 0$$

has non-trivial solution.

Sol :- For non-trivial sol. $|A| = 0 \Rightarrow k = 0, \frac{9}{2} \quad \text{Ans}$

Que (29) :- Find for what values of λ and μ the system of eq^s

$$x + y + z = 6, \quad x + 2y + 5z = 10, \quad 2x + 3y + \lambda z = \mu$$

has (i) Unique sol (ii) No sol (iii) Infinite sol.

(iv) Also find the sol for $\lambda = 2$ and $\mu = 8$.

Sol :-

$$C \cong [A:B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 5 & : & 10 \\ 2 & 3 & \lambda & : & \mu \end{bmatrix} \xrightarrow{R_2 - R_1, R_3 - 2R_1} \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 4 & : & 4 \\ 0 & 1 & \lambda - 2 & : & \mu - 12 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 4 & : & 4 \\ 0 & 0 & \lambda - 6 & : & \mu - 16 \end{bmatrix}$$

(i) Unique sol $\rightarrow \rho(A) = \rho(C) = 3$, i.e. $\lambda - 6 \neq 0$ & μ can take any value.

(ii) No sol $\rightarrow \rho(A) \neq \rho(C)$, i.e. $\lambda - 6 = 0$ & $\mu - 16 \neq 0$

(iii) Infinite sol $\rightarrow \rho(A) = \rho(C) = 2$, i.e. $\lambda - 6 = 0$ & $\mu - 16 = 0$.

(iv) putting $\lambda = 2$ & $\mu = 8$, hence

$$A \cdot X = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ -8 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x + y + z &= 6 \\ y + 4z &= 4 \\ -4z &= -8 \Rightarrow z = 2 \end{aligned}$$

Linearly Dependence and independence Vectors:

If the vectors (matrices) $X_1, X_2, \dots, X_n$, there exist a scalars $\lambda_1, \lambda_2, \dots, \lambda_n$, such that

$$\lambda_1 X_1 + \lambda_2 X_2 + \dots + \lambda_n X_n = 0, \text{ is said to be}$$

(i) linearly dependent if at least one $\lambda \neq 0$

(ii) linearly independent if all λ 's are zero.

Que (30) :- Examine the following vectors for linear dependence and find the relation if it exists.

$$X_1 = (1, 2, 4), X_2 = (2, -1, 3), X_3 = (0, 1, 2), X_4 = (-3, 7, 2)$$

Sol :- The equation $\lambda_1 X_1 + \lambda_2 X_2 + \lambda_3 X_3 + \lambda_4 X_4 = 0$

$$\Rightarrow \lambda_1 (1, 2, 4) + \lambda_2 (2, -1, 3) + \lambda_3 (0, 1, 2) + \lambda_4 (-3, 7, 2) = 0$$

$$\text{or } \lambda_1 + 2\lambda_2 + 0 \cdot \lambda_3 - 3\lambda_4 = 0$$

$$2\lambda_1 - \lambda_2 + \lambda_3 + 7\lambda_4 = 0$$

$$4\lambda_1 + 3\lambda_2 + 2\lambda_3 + 2\lambda_4 = 0$$

$$A \lambda = 0$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & -5 & 2 & 14 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} \lambda_1 + 2\lambda_2 - 3\lambda_4 &= 0 \\ -5\lambda_2 + \lambda_3 + 13\lambda_4 &= 0 \\ \lambda_3 + \lambda_4 &= 0 \end{aligned}$$

$$\text{Let } \lambda_4 = k, \text{ hence } \lambda_3 = -k$$

$$\text{and } \lambda_2 = \frac{12k}{5}, \text{ and } \lambda_1 = \frac{-9k}{5}$$

ie. given vectors are linearly dependent.

$$\text{Now from eq (1)} \quad -\frac{9k}{5} X_1 + \frac{12k}{5} X_2 - k X_3 + k X_4 = 0$$

Que (31) :- Is the system of vectors $x_1 = (2, 2, 1)^T$, $x_2 = (1, 3, 1)^T$

$x_3 = (1, 2, 2)^T$ linearly dependent?

Sol: $x_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

Now consider $\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 = 0$

$$\Rightarrow \lambda_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = 0$$

$$\Rightarrow \begin{aligned} 2\lambda_1 + \lambda_2 + \lambda_3 &= 0 \\ 2\lambda_1 + 3\lambda_2 + 2\lambda_3 &= 0 \\ \lambda_1 + \lambda_2 + 2\lambda_3 &= 0 \end{aligned}$$

which is a homogeneous eqⁿ.

Now $A\lambda = 0$

$$\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_1 \leftrightarrow R_3$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 2R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & -1 & -3 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \lambda_1 + \lambda_2 + 2\lambda_3 = 0, \quad \lambda_2 - 2\lambda_3 = 0, \quad -5\lambda_3 = 0 \Rightarrow \lambda_3 = 0$$

$$\therefore \lambda_2 = 0 \text{ \& } \lambda_1 = 0.$$

i.e. all λ 's are zero. Hence vectors are linearly independent.

Que (32) :- Find the value of λ for which the vectors $(1, -2, \lambda)$, $(2, -1, 5)$ and $(3, -5, 7\lambda)$ are linearly dependent.

Sol: Let $x_1 = (1, -2, \lambda)$, $x_2 = (2, -1, 5)$ and $x_3 = (3, -5, 7\lambda)$, there scalars a_1, a_2 and a_3 s.t.

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0$$

$$\text{or } a_1 (1, -2, \lambda) + a_2 (2, -1, 5) + a_3 (3, -5, 7\lambda) = 0$$

$$\Rightarrow \begin{aligned} a_1 + 2a_2 + 3a_3 &= 0 \\ -2a_1 - a_2 - 5a_3 &= 0 \\ \lambda a_1 + 5a_2 - 7\lambda a_3 &= 0 \end{aligned}$$

$$\lambda a_1 + 5a_2 - 7\lambda a_3 = 0$$

$$\lambda a_1 + 5a_2 - 7\lambda a_3 = 0$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -2 & -1 & -5 \\ \lambda & 5 & -7\lambda \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 \rightarrow R_2 + 2R_1$, $R_3 \rightarrow R_3 - \lambda R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 0 & 5-2\lambda & 4\lambda \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_3 \rightarrow 3R_3 - (5-2\lambda)R_2$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 0 & 0 & 14\lambda-5 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda a_1 + 5a_2 - 7\lambda a_3 = 0$$

Que (33) :- If $x_1 = (3, 1, -4)$, $x_2 = (2, 2, -3)$, $x_3 = (0, -4, 1)$, then show that

- (i) The vectors x_1 and x_2 are linearly independent.
- (ii) The vectors x_1 , x_2 and x_3 are linearly dependent.

Que (34) :- Examine the following vectors for linearly dependence and find the relation between them, if possible -

- (i) $x_1 = (1, 1, -1, 1)$, $x_2 = (1, -1, -2, -1)$, $x_3 = (3, 1, 0, 1)$
- (ii) $x_1 = (1, -1, 1)$, $x_2 = (2, 1, 1)$, $x_3 = (3, 0, 2)$
- (iii) $x_1 = (1, 0, 2, 1)$, $x_2 = (3, 1, 2, 1)$, $x_3 = (4, 6, 2, -4)$, $x_4 = (-6, 0, -3, -4)$
- (iv) $x_1 = (2, 5, 2, -3)$, $x_2 = (3, 6, 5, 2)$, $x_3 = (4, 5, 14, 14)$, $x_4 = (5, 10, 0, 4)$
- (v) $x_1 = (1, 2, 3)$, $x_2 = (2, -2, 6)$
- (vi) $x_1 = (3, 2, 4)$, $x_2 = (1, 0, 2)$, $x_3 = (1, -1, -1)$

- Ans :-
- (i) linearly dependent, $2x_1 + x_2 = x_3$
 - (ii) " " , $x_1 + x_2 = x_3$
 - (iii) " " , $2x_1 - 6x_2 + x_3 - 2x_4 = 0$
 - (iv) linearly independent.
 - (v) " "
 - (vi) " "

Characteristic Polynomial; — The determinant $|A - \lambda I|$ is known as characteristic polynomial.

Characteristic Equation; — The equation $|A - \lambda I| = 0$ is known as characteristic eqn.

Characteristic Roots OR Eigen Values; — The roots of $|A - \lambda I| = 0$ is called eigen values.

Some Properties of Eigenvalues :—

- ① Any square matrix A and its transpose A' have same eigenvalues.
- ② The sum of the eigenvalues equals to the trace of the matrix.
(Trace means the sum of the elements on principal diagonal)
- ③ The product of the eigenvalues of the matrix A equal to the determinant of A .
- ④ If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigen values of A , then the eigenvalues of
 - (i) kA are $k\lambda_1, k\lambda_2, \dots, k\lambda_n$
 - (ii) A^m are $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$
 - (iii) A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$

Que (35) :— Show that for any square matrix A , the product of all the eigen values of A equal to $|A|$, and the sum of all the eigen values of A equals to the sum of the diagonal elements.

Sol :— let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

characteristic eqn

$$\therefore |A - \lambda I| = 0 \Rightarrow \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = 0 \Rightarrow (a-\lambda)(d-\lambda) - bc = 0$$

$$\text{or } \lambda^2 - \lambda(a+d) + ad - bc = 0$$

$$\Rightarrow \lambda_1 + \lambda_2 = a + d = \text{sum of principal diagonals}$$

$$\& \lambda_1 \cdot \lambda_2 = ad - bc = |A|$$

$$ax^2 + bx + c = 0$$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha \cdot \beta = \frac{c}{a}$$

Cayley - Hamilton Theorem :- Every square matrix satisfies its own characteristic equation.

If $|A - \lambda I| = (-1)^n (\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n) = 0$, be characteristic equation, (where A is $n \times n$ matrix) then

$$A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = 0.$$

Proof :- Since the element of $(A - \lambda I)$ are at most of the first degree in λ .
hence " " $\text{adj}(A - \lambda I)$ " " " " $(n-1)$ " λ .

$$\therefore \text{Adj}(A - \lambda I) = B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-1}$$

where $B_0, B_1, B_2, \dots, B_{n-1}$ are $n \times n$ matrices.

We know that $A^{-1} = \frac{\text{adj}(A)}{|A|} \Rightarrow A A^{-1} = \frac{A \text{adj}(A)}{|A|} \Rightarrow \boxed{|A| I = A \text{adj}(A)}$

$$\therefore (A - \lambda I) \text{adj}(A - \lambda I) = |A - \lambda I| I$$

$$\text{or } (A - \lambda I) (B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-1}) = (-1)^n (\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n) I$$

equating the coefficients of same power of λ on both sides,

Coeff of $\lambda^n \rightarrow$	$-I B_0$	$= (-1)^n I$	① $\times A^n$
$\lambda^{n-1} \rightarrow$	$A B_0 - I B_1$	$= (-1)^n a_1 I$	② $\times A^{n-1}$
$\lambda^{n-2} \rightarrow$	$A B_1 - I B_2$	$= (-1)^n a_2 I$	③ $\times A^{n-2}$
	$\vdots$	$\vdots$	
$\lambda^0 \rightarrow$	$A B_{n-1}$	$= (-1)^n a_n I$	④ $\times I$

On multiplying by $A^n, A^{n-1}, A^{n-2}, \dots, I$ in eqⁿ ①, ②, ③, ④ and adding.

$$0 = (-1)^n [A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I]$$

$$\text{or } \Rightarrow A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = 0$$

Que :- Verify Cayley Hamilton Theorem and find A^{-1} , where $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$.

Sol :- Characteristic eqⁿ $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 2 & -1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 5 = 0$

$\therefore$ By C.H.T. $\boxed{A^2 - 5I = 0}$

Now $A^2 = A \cdot A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$

$\therefore A^2 - 5I = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$ i.e. C.H.T verified.

(17)

Que (36) :- Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ and hence find A^{-1} .

Sol :- Characteristic eqⁿ $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & -2 \\ 1 & 1-\lambda & 1 \\ 1 & 3 & -1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^3 - \lambda^2 - 4\lambda + 4 = 0$$

By Cayley-Hamilton Theorem $A^3 - A^2 - 4A + 4I = 0$
or $A^2 - A - 4I + 4A^{-1} = 0$

$$\text{or } A^{-1} = \frac{1}{4} (-A^2 + A + 4I)$$

$$= \frac{1}{4} \left(- \begin{bmatrix} 1 & -2 & 2 \\ 3 & 6 & -2 \\ 3 & 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \right) = \frac{1}{4} \begin{bmatrix} 4 & 4 & -4 \\ -2 & -3 & 3 \\ -2 & 1 & 1 \end{bmatrix}$$

Que (37) :- Find the characteristic eqⁿ of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ verify Cayley-Hamilton theorem and hence evaluate the matrix

$$\text{eqⁿ } - A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

Sol :- The characteristic eqⁿ $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 0 & 1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By C.H.T. $A^3 - 5A^2 + 7A - 3I = 0$ ——— (1)

$$\text{Now } A^2 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \quad A^3 = \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix}$$

$$\therefore A^3 - 5A^2 + 7A - 3I = \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix} - 5 \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + 7 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

i.e. Cayley-Hamilton Theorem is verified.

$$\text{Now } A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

$$= A^5 (A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I$$

$$= A^5 (0) + A(0) + A^2 + A + I$$

$$= 0 + 0 + A^2 + A + I$$

$$= A^2 + A + I = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Que (38) Find the characteristic eqⁿ of the matrix $A = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$ and hence find A^{-1} .

Sol :- Ch. eqⁿ $\rightarrow \lambda^3 - 6\lambda^2 + 6\lambda - 11 = 0$

$$\& A^{-1} = \frac{1}{11} \begin{bmatrix} 5 & -1 & -7 \\ -4 & 3 & 10 \\ 3 & -5 & -2 \end{bmatrix}$$

Que (39) Using Cayley-Hamilton theorem obtain the inverse of the matrix

(i) $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$

Ans :- (i) $\frac{1}{8} \begin{bmatrix} 24 & 8 & 12 \\ -10 & -2 & -6 \\ -2 & -2 & -2 \end{bmatrix}$, (ii)

Que (40) :- Find adj^t A by using Cayley-Hamilton theorem where $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$.

Ans :- $\begin{bmatrix} 0 & -3 & -3 \\ -3 & -2 & 1 \\ -3 & 7 & 1 \end{bmatrix}$

Que (41) :- Find the characteristic eqⁿ of $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, and verify that it is satisfied by A, and hence obtain A^{-1} , and also find the value of the matrix eqⁿ $A^5 - 5A^4 + 3A^3 + 6A^2 - 6A + 2I$.

Ans :- Ch. eqⁿ $\rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$

By C.H.T $\rightarrow A^3 - 6A^2 + 9A - 4I = 0$

Now $A^2 = A \cdot A = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$, $A^3 = A \cdot A^2 = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$

$$\begin{aligned} & A^5 - 5A^4 + 3A^3 + 6A^2 - 6A + 2I \\ &= A^2(A^3 - 6A^2 + 9A - 4I) + A^4 - 6A^3 + 10A^2 - 6A + 2I \\ &= A^2(0) + A(A^3 - 6A^2 + 9A - 4I) + A^2 - 2A + 2I \end{aligned}$$

$= A^2 - 2A + 2I$

$= \begin{bmatrix} 4 & -3 & 3 \\ -3 & 4 & -3 \end{bmatrix}$

Eigen vectors -
 Que 42 - Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$

Sol:- The ch. eqⁿ $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = 0 \Rightarrow \lambda = -1, -6$
 λ_1, λ_2

Now $(A - \lambda_1 I) X_1 = 0$

$\Rightarrow \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$R_2 \rightarrow R_2 + 2R_1$
 $\begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow -4x_1 + 2y_1 = 0$ or $-2x_1 + y_1 = 0$
 let $x_1 = k$, $\therefore y_1 = 2k$

hence $X_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} k \\ 2k \end{bmatrix} = k \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $R_2 \rightarrow R_2 - 2R_1$

Again $(A - \lambda_2 I) X_2 = 0 \Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow x_2 + 2y_2 = 0$, let $x_2 = k$, $y_2 = -2k$

hence $X_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} k \\ -2k \end{bmatrix} = k \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Ans

Que 43 - Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

Sol:- ch. eqⁿ $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$
 or $\lambda = 1, 2, 3$
 $\lambda_1, \lambda_2, \lambda_3$

To find the eigenvectors for the corresponding eigenvalues, we consider the matrix eqⁿ $(A - \lambda I) X = 0$

(20)

① For $\lambda = \lambda_1 = 1$

$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -z_1 = 0 \\ x_1 + y_1 + z_1 = 0 \\ \therefore x_1 + y_1 = 0 \end{matrix}$
 let $x_1 = k$, $\therefore y_1 = -k$

② For $\lambda = \lambda_2 = 2$

$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -x_2 - z_2 = 0 \Rightarrow x_2 + z_2 = 0 \\ 2y_2 - z_2 = 0 \Rightarrow x_2 + 2y_2 = 0 \\ \therefore x_2 = -2y_2 \end{matrix}$
 let $y_2 = k$, $\therefore x_2 = -2k$, $z_2 = 2k$

③ For $\lambda = \lambda_3 = 3$

$\begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -2x_3 - z_3 = 0 \\ -2y_3 + z_3 = 0 \\ x_3 + y_3 = 0 \end{matrix}$
 let $x_3 = k$, $y_3 = -k$

Que 44 :- Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Sol :- The Ch. eqⁿ $\rightarrow |A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$
 $\Rightarrow \lambda = 2, 2, 8$
 $\lambda_1, \lambda_2, \lambda_3$

Now $(A - \lambda_1 I) X_1 = 0$

$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} R_1 \rightarrow R_1 + 2R_2 \\ R_3 \rightarrow R_3 + R_2 \end{matrix} \Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ -2 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$
 $-2x_1 + y_1 - z_1 = 0$
 let $z_1 = k_1$, $y_1 = k_2$
 $\therefore x_1 = \frac{k_1 - k_2}{2}$

$\therefore X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} \frac{k_1 - k_2}{2} \\ k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} k_1 - k_2 \\ 2k_1 \\ 2k_2 \end{bmatrix}$

The required eigenvalues for corresponding to $\lambda_1 = 2 = \lambda_2$

$X_1 = \begin{bmatrix} k_1 - k_2 \\ 2k_1 \\ 2k_2 \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$

Now $(A - \lambda_3 I) X_3 = 0$

$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 + R_3 \end{matrix} \Rightarrow \begin{bmatrix} 0 & -3 & -3 \\ 0 & -6 & -6 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$
 $\begin{bmatrix} 0 & 0 & 0 \\ 0 & -6 & -6 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$
 $-6y_3 - 6z_3 = 0 \Rightarrow y_3 + z_3 = 0$ (2)

or $2x_3 - y_3 - 5z_3 = 0$ (1) $-6y_3 - 6z_3 = 0 \Rightarrow y_3 + z_3 = 0$ (2)

(1) + (2) $\Rightarrow 2x_3 - 4z_3 = 0 \Rightarrow x_3 = 2z_3$

let $z_3 = k$, $\therefore x_3 = 2k$, $y_3 = -k$

$\therefore X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 2k \\ -k \\ k \end{bmatrix} = k \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

Que 45 Find eigenvalues and eigenvectors of the following matrices

(i) $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$, (ii) $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

(iii) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(iv) $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$

(v) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

Ans :- (i) $6, -1, \begin{bmatrix} 2 \\ -5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (ii) $2, 3, 5, \begin{bmatrix} k \\ -k \\ 0 \end{bmatrix}, \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3k \\ 2k \\ k \end{bmatrix}$ (iii) $1, 1, 3, \begin{bmatrix} k_1 \\ k_2 \\ -k - k_2 \end{bmatrix}, k \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

(iv) $1, 1, 4, \begin{bmatrix} k_1 \\ k_2 \\ k_2 - k \end{bmatrix}, \begin{bmatrix} k \\ -k \\ k \end{bmatrix}$

(v) $0, 3, 5, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

Diagonalisation :-

Que (46) :- Find the eigen value and eigen vector of the modal matrix and diagonalise the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$.

Sol.:- The Char eq $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 3-\lambda & -1 \\ 0 & -1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda = 1, 2, 4$

When $\lambda = \lambda_1 = 1$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} 2y_1 - z_1 = 0, & 3z_1 = 0 \\ \therefore z_1 = 0 & \Rightarrow z_1 = 0 \end{matrix}$$

$$\therefore X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

When $\lambda = 2$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

$$\Rightarrow -x_2 = 0 \text{ \& } y_2 - z_2 = 0$$

Let $z_2 = k, \therefore y_2 = k$

$$\therefore X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

When $\lambda = 4$

$$\begin{bmatrix} -3 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -3 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -3x_3 = 0, \quad +y_3 + z_3 = 0$$

Let $y_3 = k, \therefore z_3 = -k$

$$\therefore X_3 = \begin{bmatrix} 0 \\ k \\ -k \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$\therefore$ Modal Matrix $= [X_1 \ X_2 \ X_3] = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = P$

Now $P^{-1}AP = -\frac{1}{2} \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

$$= -\frac{1}{2} \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 2 & -4 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} -2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

or $PAP = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$

Ans

(22)

Que (47) Find modal matrix and where (i) $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

also verify $P^{-1}AP = D$ (diagonal matrix)

(ii) $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

Ans :- (i) $\lambda = 2, 5$

$P = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, P^{-1}AP = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$

(ii) $\lambda = 5, -3, -3$

$P = \begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, P^{-1} = \frac{1}{8} \begin{bmatrix} -1 & -2 & 3 \\ 2 & -4 & -6 \\ 1 & 2 & 5 \end{bmatrix}$